

TUẦN 2: JSP Basics - Tài Liệu Lý Thuyết Chi Tiết

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Nội dung Tuần 2

- **Session 8:** JSP Introduction
 - **Session 9:** Implicit Objects
 - **Session 10:** JSP Include
 - **Session 11:** JSP Expression Language (EL)
 - **Session 12:** JSTL Basics
 - **Session 13:** Form Handling
 - **Session 14:** Mini Project
-

SESSION 8: JSP Introduction

8.1 JSP là gì?

JSP = **J**ava **S**erver **P**ages

Định nghĩa

JSP là một công nghệ cho phép bạn viết **HTML kết hợp với Java code** để tạo trang web động.

Khác nhau giữa Servlet và JSP:

Servlet:

Java → Biên dịch → Bytecode → Chạy
(Code Java với HTML print)

JSP:

HTML + Java Tags → Biên dịch → Servlet → Chạy
(HTML với Java code nhúng)

Ví dụ đơn giản

Servlet:

```
protected void doGet(HttpServletRequest request,
                      HttpServletResponse response) {
    response.setContentType("text/html; charset=UTF-8");
    PrintWriter out = response.getWriter();
```

```
out.println("<h1>Hello " + request.getParameter("name") + "</h1>");  
}
```

JSP (đơn giản hơn):

```
<h1>Hello <%= request.getParameter("name") %></h1>
```

8.2 Quy trình Biên dịch JSP

1. Client gửi request đến JSP
↓
2. Tomcat kiểm tra: JSP file có tồn tại chưa?
↓
3. JSP → Tomcat biên dịch thành Servlet (.java)
↓
4. Servlet → Biên dịch thành Bytecode (.class)
↓
5. Bytecode → Chạy
↓
6. Kết quả HTML → gửi về client
↓
7. Browser nhận và hiển thị

Ví dụ Quá trình Biên dịch

File JSP gốc: `hello.jsp`

```
<h1>Hello <%= new java.util.Date() %></h1>
```

Servlet được tạo: `hello_jsp.java`

```
public class hello_jsp extends HttpJspBase {  
    public void _jspService(HttpServletRequest request,  
                           HttpServletResponse response) {  
        // ... Tomcat tạo code tự động  
        out.println("<h1>Hello ");  
        out.print(new java.util.Date());  
        out.println("</h1>");  
    }  
}
```

Bytecode: `hello_jsp.class`

8.3 Bốn yếu tố chính của JSP

1. Directive - `<%@ ... %>`

Mục đích: Báo cho Tomcat hướng dẫn về JSP.

Các loại Directive:

a) Page Directive

```
<%@ page attribute="value" %>
```

Ví dụ:

```
<%@ page language="java" %>
<%@ page contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.*,java.io.*" %>
<%@ page errorPage="error.jsp" %>
<%@ page isErrorPage="true" %>
```

b) Include Directive

```
<%@ include file="header.jsp" %>
<!-- Nhúng nội dung file khác vào lúc biên dịch -->
```

c) Taglib Directive

```
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<!-- Import thư viện JSTL -->
```

Ví dụ thực tế:

```
<%@ page language="java" %>
<%@ page contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date, java.util.List" %>
<%@ page errorPage="error.jsp" %>

<html>
<head><title>Page Directive Demo</title></head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

2. Scriptlet - `<% ... %>`

Mục đích: Nhúng Java code trực tiếp vào JSP.

Cú pháp:

```
<% Java code here %>
```

Ví dụ:

```
<%
String name = request.getParameter("name");
int age = 25;

if (name != null && !name.isEmpty()) {
    out.println("<h1>Hello " + name + "</h1>");
} else {
    out.println("<h1>Please enter your name</h1>");
}
%>
```

Khi nào dùng Scriptlet:

- Viết logic phức tạp
- Loop, if/else
- Khai báo biến tạm thời
- Gọi method từ Java class

Lưu ý: Nếu quá nhiều Scriptlet, code sẽ khó đọc. Nên dùng JSP Model 2 (Servlet + JSP).

3. Expression - `<%= ... %>`

Mục đích: In giá trị của biến hoặc expression ra trang web.

Cú pháp:

```
<%= expression %>
```

Ví dụ:

```
<!-- In giá trị -->
<p>Tổng: <%= 5 + 3 %></p>

<!-- In biến -->
<% String name = "Nhân"; %>
```

```
<p>Xin chào <%= name %></p>

<!-- In kết quả method -->
<p>Thời gian hiện tại: <%= new java.util.Date() %></p>

<!-- In request parameter -->
<p>Email: <%= request.getParameter("email") %></p>
```

Tương đương với:

```
<% out.print(expression); %>
```

Lưu ý: Expression không chứa dấu chấm phẩy (🙄) ở cuối.

4. Declaration - <%! ... %>

Mục đích: Khai báo biến hoặc method ở mức class (không phải local).

Cú pháp:

```
<%!
    declaration
%>
```

Ví dụ:

```
<%!
    // Biến instance (toàn bộ JSP có thể truy cập)
    int counter = 0;
    String appName = "My JSP App";

    // Method
    public int add(int a, int b) {
        return a + b;
    }

    public String getGreeting(String name) {
        return "Hello " + name + "!";
    }
%>

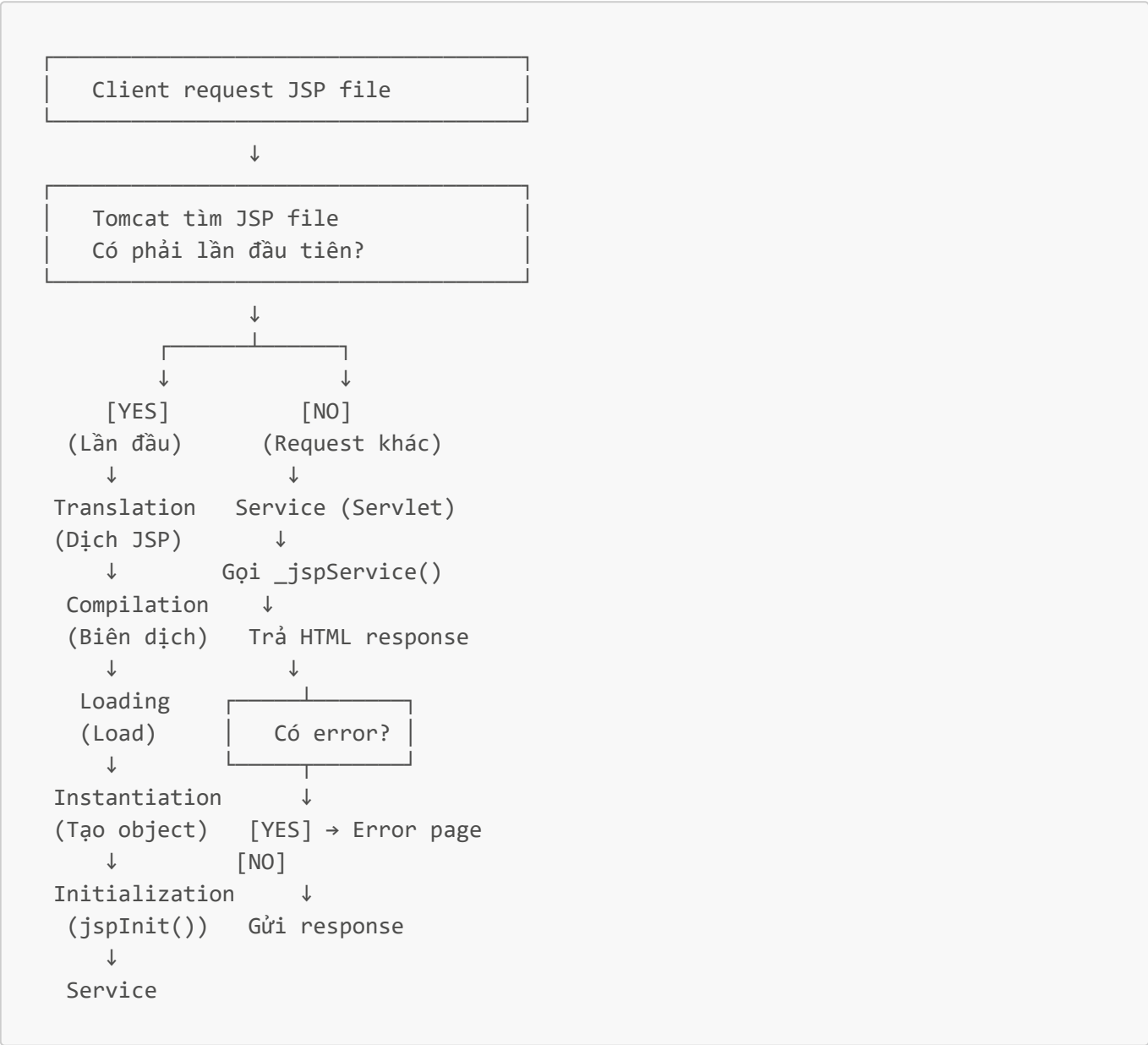
<h1><%= getGreeting("Nhân") %></h1>
<p>Tổng 5 + 3 = <%= add(5, 3) %></p>
<p>Tên ứng dụng: <%= appName %></p>
```

Khác biệt:

Scriptlet:	<% int x = 5; %>	(Local variable)
Declaration:	<%! int x = 5; %>	(Instance variable/method)

8.4 JSP Lifecycle (Vòng đời JSP)

Sơ đồ chi tiết



5 Giai đoạn JSP Lifecycle

1. Translation

hello.jsp → Tomcat biên dịch → hello_jsp.java

2. Compilation

```
hello_jsp.java → Compiler → hello_jsp.class
```

3. Loading

```
hello_jsp.class được load vào memory
```

4. Instantiation

```
Tạo object từ class: hello_jsp servlet = new hello_jsp();
```

5. Initialization

```
Gọi jspInit() - khởi tạo servlet
```

6. Service

```
Mỗi lần có request → Gọi _jspService()
```

7. Destruction

```
Khi Tomcat shutdown → Gọi jspDestroy()
```

8.5 Ví dụ JSP Hoàn Chỉnh

```
<%@ page language="java" %>
<%@ page contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date, java.text.SimpleDateFormat" %>

<%!
    // Declaration - Biến class
    private int visitCount = 0;

    // Declaration - Method
    public String formatDate(Date date) {
```

```

        SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy HH:mm:ss");
        return sdf.format(date);
    }
%>

<html>
<head>
    <title>JSP Demo</title>
</head>
<body>
    <h1>JSP Basics Demo</h1>

    <!-- Scriptlet - Java code -->
    <%
        String userName = request.getParameter("name");
        if (userName == null) {
            userName = "Visitor";
        }
        visitCount++;
    %>

    <!-- Expression - In giá trị -->
    <h2>Hello <%= userName %>! 🙌</h2>

    <!-- Sử dụng method đã khai báo -->
    <p>Thời gian hiện tại: <%= formatDate(new Date()) %></p>

    <!-- In biến class -->
    <p>Lượt truy cập: <%= visitCount %></p>

    <!-- Scriptlet - Loop -->
    <h3>Danh sách số:</h3>
    <ul>
    <%
        for (int i = 1; i <= 5; i++) {
            out.println("<li>Số " + i + "</li>");
        }
    %>
    </ul>
</body>
</html>

```

Tóm tắt Session 8

Yếu tố	Cú pháp	Mục đích
Directive	<%@ ... %>	Hướng dẫn cho Tomcat
Scriptlet	<% ... %>	Nhúng Java code
Expression	<%= ... %>	In giá trị

Yếu tố	Cú pháp	Mục đích
Declaration	<%! ... %>	Khai báo biến/method class

SESSION 9: Implicit Objects

9.1 Implicit Objects là gì?

Implicit Objects = Các object được Tomcat tự động cung cấp trong mỗi JSP.

Bạn không cần phải tạo, có sẵn để dùng.

JSP Implicit Objects
1. request
2. response
3. session
4. application
5. out
6. config
7. pageContext
8. page
9. exception

9.2 Request Object

Loại: `javax.servlet.http.HttpServletRequest`

Mục đích: Lấy thông tin về HTTP request từ client.

Các phương thức thường dùng

```
<%
    // Lấy parameter từ form hoặc query string
    String name = request.getParameter("name");
    String[] hobbies = request.getParameterValues("hobby");

    // Lấy thông tin request
    String method = request.getMethod();           // GET, POST
    String uri = request.getRequestURI();          // /app/login
    String query = request.getQueryString();       // name=john&age=25
    String host = request.getServerName();         // localhost
    int port = request.getServerPort();           // 8080
%>
```

```
// Lấy header
String userAgent = request.getHeader("User-Agent");
String acceptLang = request.getHeader("Accept-Language");

// Lấy cookie
Cookie[] cookies = request.getCookies();

// Lấy Client IP
String clientIP = request.getRemoteAddr();
String clientHost = request.getRemoteHost();

// Lấy session
HttpSession session = request.getSession();

// Lấy session(false) - không tạo mới nếu không có
HttpSession session2 = request.getSession(false);
%>

<h2>Request Information</h2>
<p>Method: <%= request.getMethod() %></p>
<p>URI: <%= request.getRequestURI() %></p>
<p>Name: <%= request.getParameter("name") %></p>
<p>Client IP: <%= request.getRemoteAddr() %></p>
```

9.3 Response Object

Loại: `javax.servlet.http.HttpServletResponse`

Mục đích: Gửi dữ liệu response về cho client.

Các phương thức thường dùng

```
<%
// Set content type (cần set ở đầu JSP)
response.setContentType("text/html; charset=UTF-8");

// Set status
response.setStatus(HttpServletResponse.SC_OK);
response.setStatus(HttpServletResponse.SC_NOT_FOUND);

// Gửi error
response.sendError(HttpServletResponse.SC_NOT_FOUND, "Page not found");

// Redirect
response.sendRedirect("https://example.com");
response.sendRedirect("/app/home");

// Set header
response.setHeader("Content-Disposition", "attachment; filename=file.pdf");
response.addHeader("Custom-Header", "value");
```

```
// Set cookie
Cookie cookie = new Cookie("username", "john");
response.addCookie(cookie);
%>

<%
// Ví dụ: Redirect nếu không có quyền
if (!isAdmin) {
    response.sendRedirect("/app/unauthorized");
}
%>
```

9.4 Session Object

Loại: `javax.servlet.http.HttpSession`

Mục đích: Lưu thông tin session của client.

Các phương thức thường dùng

```
<%
// Lấy hoặc tạo session
HttpSession sess = request.getSession();

// Set attribute
sess.setAttribute("username", "john");
sess.setAttribute("role", "admin");

// Get attribute
String username = (String) sess.getAttribute("username");

// Xóa specific attribute
sess.removeAttribute("username");

// Xóa toàn bộ session
sess.invalidate();

// Set timeout (phút)
sess.setMaxInactiveInterval(30);

// Lấy session ID
String sessionId = sess.getId();

// Lấy creation time
long createTime = sess.getCreationTime();
%>

<%
// Ví dụ: Check login
```

```
String user = (String) session.getAttribute("username");
if (user == null) {
    response.sendRedirect("/app/login");
    return;
}

%>

<h1>Hello <%= user %></h1>
```

9.5 Application Object

Loại: `javax.servlet.ServletContext`

Mục đích: Lưu thông tin toàn ứng dụng (tất cả client).

Application Scope
(Shared by all clients)

Client A ↔ Client B ↔ Client C

Các phương thức thường dùng

```
<%
// Set attribute (toàn ứng dụng)
application.setAttribute("appName", "My App");
application.setAttribute("totalUsers", 100);

// Get attribute
String appName = (String) application.getAttribute("appName");
Integer totalUsers = (Integer) application.getAttribute("totalUsers");

// Remove attribute
application.removeAttribute("totalUsers");

// Lấy real path
String realPath = application.getRealPath("/uploads/");

// Lấy init parameter
String dbUrl = application.getInitParameter("db.url");
%>

<%
// Ví dụ: Đếm lượt truy cập toàn app
Integer visitCount = (Integer) application.getAttribute("visitCount");
if (visitCount == null) {
    visitCount = 1;
}
```

```
    } else {  
        visitCount++;  
    }  
    application.setAttribute("visitCount", visitCount);  
%>  
  
<p>Tổng lượt truy cập: <%= application.getAttribute("visitCount") %></p>
```

9.6 Out Object

Loại: `javax.servlet.jsp.JspWriter`

Mục đích: In dữ liệu ra trang web (tương tự `PrintWriter`).

Các phương thức

```
<%  
    // Print text  
    out.print("Hello");  
    out.println("World");  
  
    // Print variable  
    String name = "Nhân";  
    out.print(name);  
  
    // Out.println thêm dòng mới  
    out.println("<h1>Heading</h1>");  
    out.println("<p>Paragraph</p>");  
%>  
  
<!-- Tương đương -->  
<h1>Heading</h1>  
<p>Paragraph</p>  
  
<%  
    // Flush output  
    out.flush();  
  
    // Clear output (xóa buffer)  
    out.clearBuffer();  
  
    // Get buffer info  
    int size = out.getBufferSize();  
    int remain = out.getRemaining();  
%>
```

9.7 Config Object

Loại: `javax.servlet.ServletConfig`

Mục đích: Lấy thông tin config từ web.xml.

```
<%
    // Lấy servlet name
    String servletName = config.getServletName();

    // Lấy init parameter từ web.xml
    String dbUrl = config.getInitParameter("database.url");

    // Lấy servlet context
    ServletContext context = config.getServletContext();
%>
```

Ví dụ web.xml:

```
<servlet>
    <servlet-name>MyServlet</servlet-name>
    <jsp-file>/index.jsp</jsp-file>
    <init-param>
        <param-name>database.url</param-name>
        <param-value>jdbc:mysql://localhost:3306/db</param-value>
    </init-param>
</servlet>
```

Sử dụng trong JSP:

```
<%
    String dbUrl = config.getInitParameter("database.url");
%>
<p>Database URL: <%= dbUrl %></p>
```

9.8 PageContext Object

Loại: `javax.servlet.jsp.PageContext`

Mục đích: Truy cập tất cả implicit object và scope.

```
<%
    // Page scope
    pageContext.setAttribute("name", "Nhân", PageContext.PAGE_SCOPE);

    // Request scope
    pageContext.setAttribute("email", "nhan@example.com",
```

```
PageContext.REQUEST_SCOPE);

// Session scope
pageContext.setAttribute("user", "admin", PageContext.SESSION_SCOPE);

// Application scope
pageContext.setAttribute("appName", "MyApp", PageContext.APPLICATION_SCOPE);

// Get attribute từ không gian nào đó
String name = (String) pageContext.getAttribute("name");

// Lấy request/response
HttpServletRequest req = (HttpServletRequest) pageContext.getRequest();
HttpServletResponse resp = (HttpServletResponse) pageContext.getResponse();

// Forward hoặc include
pageContext.forward("other.jsp");
pageContext.include("header.jsp");
%>
```

9.9 Bảng Tóm Tắt Implicit Objects

Object	Loại	Scope	Mục đích
request	HttpServletRequest	Request	Lấy info request
response	HttpServletResponse	Page	Gửi response
session	HttpSession	Session	Lưu dữ liệu session
application	ServletContext	Application	Lưu dữ liệu toàn app
out	JspWriter	Page	In dữ liệu ra trang
config	ServletConfig	Page	Lấy config từ web.xml
pageContext	PageContext	Page	Truy cập toàn bộ scope

Tóm tắt Session 9

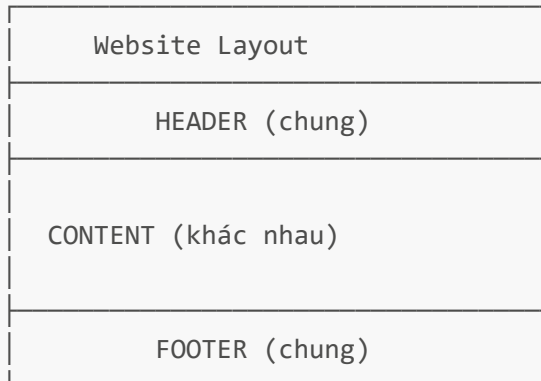
Sử dụng Implicit Objects để:

- ☒ Lấy request parameter (request)
- ☒ Lưu session data (session)
- ☒ Chia sẻ data toàn app (application)
- ☒ In output ra trang (out)
- ☒ Điều hướng (response)

SESSION 10: JSP Include

10.1 Tại sao cần Include?

Tình huống thực tế:



Nếu không dùng include, phải code header/footer ở mỗi JSP → lặp code!

Giải pháp: Include file chứa header/footer

10.2 Page Directive Include

Cú pháp:

```
<%@ include file="path/to/file.jsp" %>
```

Tính chất:

- Nhúng file lúc **biên dịch** (translation time)
- Kết hợp 2 file thành 1 servlet
- Chia sẻ biến giữa file
- Nhanh hơn (chỉ biên dịch 1 lần)

Ví dụ

header.jsp:

```
<!-- header.jsp -->
<div class="header" style="background-color: #333; color: white; padding: 20px;
text-align: center;">
  <h1>🌐 My Website</h1>
  <nav>
    <a href="/app/home" style="color: white; margin: 0 10px;">Home</a>
    <a href="/app/about" style="color: white; margin: 0 10px;">About</a>
    <a href="/app/contact" style="color: white; margin: 0 10px;">Contact</a>
  </nav>
</div>
```

footer.jsp:

```
<!-- footer.jsp -->
<div class="footer" style="background-color: #333; color: white; padding: 20px;
text-align: center;">
  <p>&copy; 2025 My Website. All rights reserved.</p>
  <p>Contact: info@example.com</p>
</div>
```

index.jsp (trang chủ):

```
<%@ include file="header.jsp" %>

<div class="content" style="padding: 20px;">
  <h2>Welcome to My Website!</h2>
  <p>This is the home page content.</p>
</div>

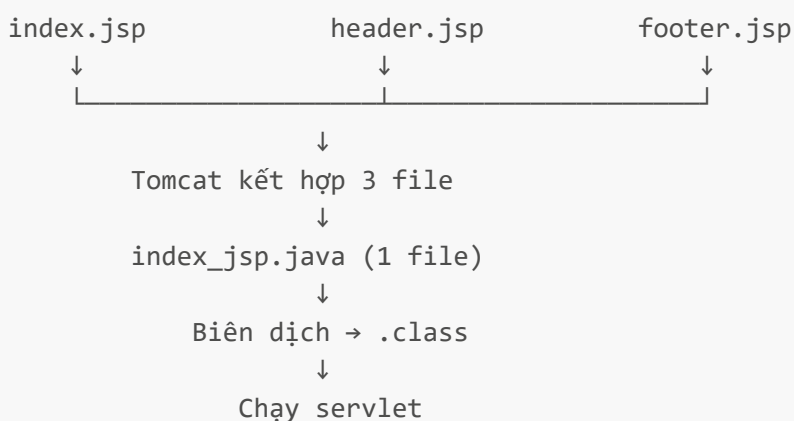
<%@ include file="footer.jsp" %>
```

about.jsp (trang riêng):

```
<%@ include file="header.jsp" %>

<div class="content" style="padding: 20px;">
  <h2>About Us</h2>
  <p>This is the about page content.</p>
</div>

<%@ include file="footer.jsp" %>
```

Quy trình Biên dịch

10.3 JSP Action Include

Cú pháp:

```
<jsp:include page="path/to/file.jsp" />
```

Tính chất:

- Nhúng file lúc **chạy** (runtime)
- Mỗi file là servlet riêng
- Không chia sẻ biến trực tiếp
- Chậm hơn (nhưng linh hoạt hơn)

Ví dụ

```
<%@ page contentType="text/html; charset=UTF-8" %>

<html>
<head><title>Dynamic Page</title></head>
<body>
    <jsp:include page="header.jsp" />

    <div class="content" style="padding: 20px;">
        <h2>Welcome!</h2>
        <p>Content here</p>
    </div>

    <jsp:include page="footer.jsp" />
</body>
</html>
```

Truyền Parameter qua Include

```
<!-- main.jsp --%>
<jsp:include page="user_card.jsp">
    <jsp:param name="username" value="john" />
    <jsp:param name="role" value="admin" />
</jsp:include>

<!-- user_card.jsp --%>
<%
    String username = request.getParameter("username");
    String role = request.getParameter("role");
%>
<div>
```

```
<p>Username: <%= username %></p>
<p>Role: <%= role %></p>
</div>
```

10.4 So sánh Include

Yếu tố	Page Directive	JSP Action
Cú pháp	<%@ include ... %>	<jsp:include ... />
Lúc nào	Biên dịch (Translation)	Chạy (Runtime)
File sinh ra	1 servlet	Nhiều servlet
Chia sẻ biến	Có (cùng servlet)	Không (servlet riêng)
Truyền parameter	Không	Có
Tốc độ	Nhanh	Chậm hơn
Linh hoạt	Ít	Nhiều
Dùng khi	Header/Footer cố định	Content động

10.5 Ví dụ Thực Tế: Website Layout

Cấu trúc:

```
webapp/
├ header.jsp
├ footer.jsp
├ sidebar.jsp
├ index.jsp
├ products.jsp
└ about.jsp
```

header.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>

<div style="background-color: #2c3e50; color: white; padding: 20px; margin-bottom: 20px;">
  <h1><img alt="book icon" data-bbox="184 858 202 872"/> Learning Portal</h1>
  <nav>
    <a href="index.jsp" style="color: white; margin-right: 20px; text-decoration: none;">Home</a>
    <a href="products.jsp" style="color: white; margin-right: 20px; text-decoration: none;">Products</a>
```

```
        <a href="about.jsp" style="color: white; text-decoration: none;">About</a>
    </nav>
</div>
```

index.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>

<!DOCTYPE html>
<html>
<head>
    <title>Home</title>
    <style>
        body { font-family: Arial, sans-serif; margin: 0; }
        .container { max-width: 1200px; margin: 0 auto; }
        .main-content { display: flex; }
        .content { flex: 1; padding: 20px; }
        .sidebar { width: 250px; background-color: #ecf0f1; padding: 20px; }
    </style>
</head>
<body>
    <div class="container">
        <%@ include file="header.jsp" %>

        <div class="main-content">
            <div class="content">
                <h2>Welcome to Home Page</h2>
                <p>This is the main content area.</p>
            </div>

            <div class="sidebar">
                <h3>Sidebar</h3>
                <ul>
                    <li><a href="#">Link 1</a></li>
                    <li><a href="#">Link 2</a></li>
                    <li><a href="#">Link 3</a></li>
                </ul>
            </div>
        </div>

        <%@ include file="footer.jsp" %>
    </div>
</body>
</html>
```

footer.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>

<div style="background-color: #2c3e50; color: white; padding: 20px; text-align: center; margin-top: 20px;">
    <p>&copy; 2025 Learning Portal. All rights reserved.</p>
</div>
```

Tóm tắt Session 10

Tính năng	Page Include	JSP Include
Dùng	Header/Footer cố định	Content động
Cú pháp	<%@ include file="..." %>	<jsp:include page="..." />
Thời điểm	Biên dịch	Runtime
Tốc độ	Nhanh	Chậm

SESSION 11: JSP Expression Language (EL)

11.1 EL là gì?

EL = **E**xpression **L**anguage

Mục đích: Cung cấp cách đơn giản hơn để truy cập dữ liệu trong JSP.

So sánh:

```
<!-- Scriptlet (cũ, phức tạp) -->
<% String name = (String) session.getAttribute("username"); %>
<h1>Hello <%= name %></h1>

<!-- EL (mới, đơn giản) -->
<h1>Hello ${sessionScope.username}</h1>
```

11.2 EL Scopes

Dữ liệu có thể lưu ở 4 scope:

1. **pageScope** - Trang hiện tại

```
<% pageContext.setAttribute("message", "Page message"); %>

${pageScope.message} <!-- In: Page message -->
```

2. **requestScope** - Request hiện tại

```
<% request.setAttribute("user", "john"); %>

${requestScope.user} <!-- In: john -->
```

3. **sessionScope** - Session hiện tại

```
<% session.setAttribute("username", "admin"); %>

${sessionScope.username} <!-- In: admin -->
```

4. **applicationScope** - Toàn ứng dụng

```
<% application.setAttribute("appName", "My App"); %>

${applicationScope.appName} <!-- In: My App -->
```

11.3 EL Automatic Lookup

Nếu không chỉ scope, EL sẽ tìm tự động:

```
pageScope → requestScope → sessionScope → applicationScope
```

Ví dụ:

```
<% pageContext.setAttribute("name", "Page"); %>
<% request.setAttribute("name", "Request"); %>
<% session.setAttribute("name", "Session"); %>

${ name } <!-- Tìm từ pageScope trước → In: Page -->
```

11.4 Truy cập Object Properties

Dạng Dot Notation

```
<!-- Truy cập property của object -->
${ user.name }
${ user.email }
${ user.address.city }

<!-- Tương đương -->
user.getName()
user.getEmail()
user.getAddress().getCity()
```

Dạng Bracket Notation

```
<!-- Truy cập array hoặc map -->
${ array[0] }
${ map['key'] }
${ map["key"] }

<!-- Tương đương -->
array[0]
map.get("key")
```

Ví dụ thực tế

```
<%
    class User {
        public String name = "Nhân";
        public int age = 25;
        public String getEmail() { return "nhan@example.com"; }
    }

    User user = new User();
    request.setAttribute("user", user);

    List<String> hobbies = Arrays.asList("Reading", "Gaming", "Coding");
    request.setAttribute("hobbies", hobbies);
%>

<p>Name: ${ user.name }</p>
<p>Age: ${ user.age }</p>
<p>Email: ${ user.email }</p>
<p>First hobby: ${ hobbies[0] }</p>
<p>Second hobby: ${ hobbies[1] }</p>
```

11.5 Request Parameters và Headers

Lấy Request Parameter

```
<!-- request.getParameter("name") -->
<p>Name: ${ param.name }</p>

<!-- request.getParameterValues("hobby") -->
<p>Hobbies: ${ paramValues.hobby[0] }, ${ paramValues.hobby[1] }</p>
```

Ví dụ:

URL: `http://localhost:8080/app?name=john&age=25&hobby=reading&hobby=coding`

```
<p>Name: ${ param.name }</p>           <!-- In: john -->
<p>Age: ${ param.age }</p>           <!-- In: 25 -->
<p>Hobby 1: ${ paramValues.hobby[0] }</p> <!-- In: reading -->
<p>Hobby 2: ${ paramValues.hobby[1] }</p> <!-- In: coding -->
```

Lấy Request Header

```
<!-- request.getHeader("User-Agent") -->
<p>User Agent: ${ header['User-Agent'] }</p>

<!-- request.getHeaderValues("Accept-Language") -->
<p>Languages: ${ headerValues['Accept-Language'][0] }</p>
```

11.6 EL Operators

Arithmetic Operators

```
${ 5 + 3 }           <!-- 8 -->
${ 10 - 2 }          <!-- 8 -->
${ 4 * 3 }           <!-- 12 -->
${ 12 / 3 }          <!-- 4 -->
${ 12 / 5 }          <!-- 2.4 -->
${ 12 % 5 }          <!-- 2 -->
```

Comparison Operators

```
${ 5 == 5 }          <!-- true -->
${ 5 != 3 }          <!-- true -->
```

```
${ 10 > 5 }           <!-- true -->
${ 5 < 10 }           <!-- true -->
${ 10 >= 5 }          <!-- true -->
${ 5 <= 10 }          <!-- true -->
```

Logical Operators

```
${ true && true }      <!-- true -->
${ true || false }     <!-- true -->
${ !true }             <!-- false -->
${ (5 > 3) && (3 > 1) } <!-- true -->
```

Empty Operator

```
${ empty user }        <!-- true if user is null/empty -->
${ not empty user }     <!-- false if user is null/empty -->
```

Ternary Operator

```
${ age > 18 ? "Adult" : "Minor" }
${ status == "active" ? "Active" : "Inactive" }
```

11.7 EL Functions

```
<!-- Length -->
${ fn:length("Hello") }           <!-- 5 -->
${ fn:length(list) }              <!-- Size of list -->

<!-- Substring -->
${ fn:substring("Hello", 0, 3) } <!-- Hel -->

<!-- Uppercase/Lowercase -->
${ fn:toUpperCase("hello") }      <!-- HELLO -->
${ fn:toLowerCase("HELLO") }      <!-- hello -->

<!-- Contains -->
${ fn:contains("Hello World", "World") } <!-- true -->

<!-- Split -->
${ fn:split("a,b,c", ",") }       <!-- Array [a, b, c] -->

<!-- Join -->
${ fn:join(array, ",") }           <!-- a,b,c -->
```

Cần import JSTL:

```
<%@ taglib prefix="fn" uri="http://java.sun.com/jsp/jstl/functions" %>
```

11.8 Ví dụ Thực Tế EL

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ taglib prefix="fn" uri="http://java.sun.com/jsp/jstl/functions" %>

<%
    // Set dữ liệu
    session.setAttribute("username", "Nhân");
    request.setAttribute("age", 25);
    request.setAttribute("email", "nhan@example.com");
    request.setAttribute("message", "    Xin chào    ");
%>

<!DOCTYPE html>
<html>
<head><title>EL Demo</title></head>
<body>
    <h1>Expression Language Examples</h1>

    <!-- Simple property access -->
    <p>Username (session): ${ sessionScope.username }</p>
    <p>Age (request): ${ requestScope.age }</p>
    <p>Email: ${ email }</p>

    <!-- Arithmetic -->
    <p>Next year age: ${ age + 1 }</p>
    <p>Age doubled: ${ age * 2 }</p>

    <!-- Comparison -->
    <p>Is adult? ${ age >= 18 ? "Yes" : "No" }</p>

    <!-- String functions -->
    <p>Email uppercase: ${ fn:toUpperCase(email) }</p>
    <p>Message length: ${ fn:length(fn:trim(message)) }</p>
    <p>Email contains '@'? ${ fn:contains(email, "@") ? "Yes" : "No" }</p>

    <!-- Request parameter -->
    <p>Search keyword: ${ param.q }</p>
    <p>Is search empty? ${ empty param.q ? "Yes" : "No" }</p>
</body>
</html>
```

Tóm tắt Session 11

Yếu tố	EL	Scriptlet
Cú pháp	<code>\${ expression }</code>	<code><% code %></code>
Đơn giản	Có	Không
Readability	Cao	Thấp
Dùng	Hiển thị dữ liệu	Logic phức tạp

SESSION 12: JSTL Basics

12.1 JSTL là gì?

JSTL = **J**ava**S**erver **P**ages **S**tandard **T**ag **L**ibrary

Mục đích: Cung cấp các tag tiêu chuẩn để thay thế Scriptlet, code JSP sạch hơn.

```
Trước: <% for (int i = 0; i < 5; i++) { %> ... <% } %>
Sau:   <c:forEach var="i" begin="0" end="4"> ... </c:forEach>
```

12.2 Tài nguyên JSTL

Thư mục Core JSTL (phổ biến nhất):

```
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
```

JAR file cần:

- `jstl-api-1.2.jar`
- `jstl-impl-1.2.jar`

Hoặc: `javax.servlet.jsp.jstl-1.2.x.jar`

12.3 Core Tags

1. `<c:set>` - Gán giá trị

```
<%-- Gán vào page scope --%>
<c:set var="name" value="Nhân" />
```

```
<!-- Gán vào object property --%>
<c:set target="${ user }" property="age" value="25" />

<!-- Gán biểu thức --%>
<c:set var="total" value="${ 5 + 3 }" />
```

2. <c:if> - Điều kiện

```
<c:if test="${ age >= 18 }">
  <p>You are an adult</p>
</c:if>

<c:if test="${ not empty username }">
  <p>Hello ${ username }</p>
</c:if>
```

3. <c:choose>, <c:when>, <c:otherwise> - Switch/Case

```
<c:choose>
  <c:when test="${ role == 'admin' }">
    <p>Welcome Admin</p>
  </c:when>
  <c:when test="${ role == 'user' }">
    <p>Welcome User</p>
  </c:when>
  <c:otherwise>
    <p>Unknown role</p>
  </c:otherwise>
</c:choose>
```

4. <c:forEach> - Loop

```
<!-- Loop từ 1 đến 5 --%>
<c:forEach var="i" begin="1" end="5">
  <p>Number: ${ i }</p>
</c:forEach>

<!-- Loop với step --%>
<c:forEach var="i" begin="0" end="10" step="2">
  <p>${ i }</p> <!-- 0, 2, 4, 6, 8, 10 -->
</c:forEach>

<!-- Loop qua collection --%>
<c:forEach var="item" items="${ items }">
  <p>${ item }</p>
</c:forEach>
```

```

<!-- Loop qua map -->
<c:forEach var="entry" items="${ map }">
    <p>Key: ${ entry.key }, Value: ${ entry.value }</p>
</c:forEach>

<!-- Loop với status -->
<c:forEach var="item" items="${ items }" varStatus="status">
    <p>
        Item #${ status.count }: ${ item }
        (First: ${ status.first }, Last: ${ status.last })
    </p>
</c:forEach>

```

5. <c:redirect> - Chuyển hướng

```

<c:redirect url="https://example.com" />
<c:redirect url="/app/home" />

```

6. <c:url> - Tạo URL

```

<c:url var="loginUrl" value="/app/login">
    <c:param name="username" value="john" />
    <c:param name="returnUrl" value="/home" />
</c:url>

<a href="${ loginUrl }">Login</a>
<!-- Sinh ra: /app/login?username=john&returnUrl=%2Fhome -->

```

7. <c:out> - In giá trị (với escape HTML)

```

<!-- Escape HTML - an toàn -->
<c:out value="${ userInput }" default="No value" />

<!-- Không escape -->
<c:out value="${ userInput }" escapeXml="false" />

```

12.4 Ví dụ Thực Tế: Danh sách sản phẩm

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>

```

```

<%
    // Tạo dữ liệu
    java.util.List<String> products = java.util.Arrays.asList(
        "Laptop", "Mouse", "Keyboard", "Monitor", "Headphones"
    );
    request.setAttribute("products", products);

    java.util.Map<String, Integer> prices = new java.util.HashMap<>();
    prices.put("Laptop", 15000000);
    prices.put("Mouse", 500000);
    prices.put("Keyboard", 1000000);
    prices.put("Monitor", 5000000);
    prices.put("Headphones", 2000000);
    request.setAttribute("prices", prices);
%>

<!DOCTYPE html>
<html>
<head>
    <title>Products</title>
    <style>
        table { border-collapse: collapse; width: 100%; }
        th, td { border: 1px solid #ddd; padding: 10px; text-align: left; }
        th { background-color: #4CAF50; color: white; }
    </style>
</head>
<body>
    <h1>📦 Product List</h1>

    <table>
        <thead>
            <tr>
                <th>No.</th>
                <th>Product Name</th>
                <th>Price (VND)</th>
                <th>Action</th>
            </tr>
        </thead>
        <tbody>
            <c:forEach var="product" items="${ products }" varStatus="status">
                <tr>
                    <td>${ status.count }</td>
                    <td>${ product }</td>
                    <td>${ prices[product] }</td>
                    <td>
                        <c:choose>
                            <c:when test="${ prices[product] > 10000000 }">
                                <span style="color: red; font-weight:
bold;">Expensive</span>
                            </c:when>
                            <c:when test="${ prices[product] > 1000000 }">
                                <span style="color: orange;">Medium</span>
                            </c:when>
                            <c:otherwise>

```

```

        <span style="color: green;">Cheap</span>
      </c:otherwise>
    </c:choose>
  </td>
</tr>
</c:forEach>
</tbody>
</table>

<p>Total products: <strong><c:out value="${ products.size() }" default="0" />
</strong></p>
</body>
</html>

```

12.5 So sánh: Scriptlet vs JSTL

```

<!-- SCRIPTLET - Code lộn xộn -->
<%
    List<String> items = (List<String>) request.getAttribute("items");
    if (items != null && items.size() > 0) {
        for (String item : items) {
            out.println("<li>" + item + "</li>");
        }
    } else {
        out.println("<p>No items</p>");
    }
%>

<!-- JSTL - Code sạch -->
<c:choose>
  <c:when test="${ not empty items }">
    <c:forEach var="item" items="${ items }">
      <li>${ item }</li>
    </c:forEach>
  </c:when>
  <c:otherwise>
    <p>No items</p>
  </c:otherwise>
</c:choose>

```

Tóm tắt Session 12

Tag	Mục đích
<c:set>	Gán giá trị
<c:if>	Điều kiện if

Tag	Mục đích
<code><c:choose></code>	Switch/case
<code><c:forEach></code>	Loop
<code><c:redirect></code>	Redirect
<code><c:url></code>	Tạo URL
<code><c:out></code>	In giá trị



SESSION 13: Form Handling

13.1 Quy trình Form Handling

```
HTML Form (index.jsp)
  ↓ (User submits)
Form POST request
  ↓ (Tomcat receives)
Servlet xử lý (Controller)
  ↓ (Get parameters)
Logic xử lý
  ↓ (Set attributes)
JSP (View) hiển thị
  ↓
Response gửi về Browser
```

13.2 HTML Form

```
<form method="POST" action="register">
  <label>Name:</label>
  <input type="text" name="name" required />

  <label>Email:</label>
  <input type="email" name="email" required />

  <label>Age:</label>
  <input type="number" name="age" />

  <label>Country:</label>
  <select name="country">
    <option value="">-- Select --</option>
    <option value="Vietnam">Vietnam</option>
    <option value="USA">USA</option>
  </select>
```

```
<label>Interests:</label>
<input type="checkbox" name="interest" value="Sports" /> Sports
<input type="checkbox" name="interest" value="Music" /> Music
<input type="checkbox" name="interest" value="Reading" /> Reading

<button type="submit">Register</button>
</form>
```

13.3 Servlet xử lý Form

```
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.IOException;

@WebServlet("/register")
public class RegisterServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {

        // 1. Lấy form data
        String name = request.getParameter("name");
        String email = request.getParameter("email");
        String ageStr = request.getParameter("age");
        String country = request.getParameter("country");
        String[] interests = request.getParameterValues("interest");

        // 2. Validate
        if (name == null || name.isEmpty()) {
            request.setAttribute("error", "Name is required");
            request.getRequestDispatcher("register.jsp").forward(request,
response);
            return;
        }

        if (email == null || !email.contains("@")) {
            request.setAttribute("error", "Invalid email");
            request.getRequestDispatcher("register.jsp").forward(request,
response);
            return;
        }

        int age = 0;
        if (ageStr != null && !ageStr.isEmpty()) {
            try {
```

```
        age = Integer.parseInt(ageStr);
    } catch (NumberFormatException e) {
        request.setAttribute("error", "Invalid age");
        request.getRequestDispatcher("register.jsp").forward(request,
response);
        return;
    }
}

// 3. Xử lý logic (lưu DB, gửi email, v.v.)
// ... code lưu DB ...

// 4. Set attributes cho JSP
request.setAttribute("success", true);
request.setAttribute("name", name);
request.setAttribute("email", email);
request.setAttribute("age", age);
request.setAttribute("country", country);
request.setAttribute("interests", interests);

// 5. Forward đến JSP
request.getRequestDispatcher("result.jsp").forward(request, response);
}
}
```

13.4 JSP Form (index.jsp)

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-
8" %>

<!DOCTYPE html>
<html>
<head>
    <title>Registration Form</title>
    <style>
        body { font-family: Arial; max-width: 600px; margin: 50px auto; }
        .form-group { margin: 15px 0; }
        label { display: block; margin-bottom: 5px; font-weight: bold; }
        input[type="text"],
        input[type="email"],
        input[type="number"],
        select { width: 100%; padding: 8px; }
        button { padding: 10px 20px; background-color: #4CAF50; color: white;
border: none; }
        .error { color: red; }
    </style>
</head>
<body>
    <h1>Registration Form</h1>
```

```

<% if (request.getAttribute("error") != null) { %>
    <p class="error">Error: <%= request.getAttribute("error") %></p>
<% } %>

<form method="POST" action="register">
    <div class="form-group">
        <label>Name:</label>
        <input type="text" name="name" required />
    </div>

    <div class="form-group">
        <label>Email:</label>
        <input type="email" name="email" required />
    </div>

    <div class="form-group">
        <label>Age:</label>
        <input type="number" name="age" />
    </div>

    <div class="form-group">
        <label>Country:</label>
        <select name="country">
            <option value="">-- Select Country --</option>
            <option value="Vietnam">Vietnam</option>
            <option value="Thailand">Thailand</option>
            <option value="USA">USA</option>
        </select>
    </div>

    <div class="form-group">
        <label>Interests:</label>
        <input type="checkbox" name="interest" value="Sports" /> Sports
        <input type="checkbox" name="interest" value="Music" /> Music
        <input type="checkbox" name="interest" value="Reading" /> Reading
    </div>

    <button type="submit">Register</button>
</form>
</body>
</html>

```

13.5 JSP Result (result.jsp)

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>

<!DOCTYPE html>
<html>

```

```

<head>
  <title>Registration Result</title>
  <style>
    body { font-family: Arial; max-width: 600px; margin: 50px auto; }
    .success { background-color: #dff0d8; padding: 15px; border-radius: 5px; }
    .info { background-color: #d9edf7; padding: 10px; margin: 10px 0; }
    a { color: blue; text-decoration: none; }
  </style>
</head>
<body>
  <h1>Registration Successful! ☒

```

13.6 MVC Flow Đầy Đủ

```
User fills form
    ↓
Submit POST
    ↓
[Servlet - Controller]
├─ Lấy parameters (Model)
├─ Validate dữ liệu
├─ Xử lý logic (business logic)
├─ Lưu dữ liệu
├─ Set attributes
    ↓ Forward
[JSP - View]
├─ Nhận dữ liệu từ request
├─ Hiển thị kết quả
├─ Gửi HTML
    ↓
Browser nhận HTML
    ↓
User thấy result
```

Tóm tắt Session 13

Quy trình Form Handling:

1. ☒ Tạo HTML Form
2. ☒ Submit POST request
3. ☒ Servlet lấy parameters
4. ☒ Validate dữ liệu
5. ☒ Xử lý logic
6. ☒ Set attributes
7. ☒ Forward đến JSP
8. ☒ JSP hiển thị kết quả

SESSION 14: Mini Project - Simple Form Submission App

14.1 Yêu cầu Dự Án

Tạo ứng dụng **Student Registration System** với:

1. **Trang chủ** - Form đăng ký sinh viên
2. **Validation** - Server-side validation
3. **Processing** - Servlet xử lý

4. **Result** - Hiển thị kết quả
 5. **Database** - Lưu vào file (text)
-

14.2 Cấu trúc Project

```
StudentRegistration/
├─ src/main/java/
│   └─ com/example/servlets/
│       └─ RegistrationServlet.java
├─ src/main/webapp/
│   └─ WEB-INF/
│       └─ web.xml
│   └─ index.jsp          (Form)
│   └─ result.jsp         (Kết quả)
│   └─ error.jsp          (Lỗi)
└─ build/
    └─ StudentRegistration.war
```

14.3 Code - RegistrationServlet.java

```
// src/main/java/com/example/servlets/RegistrationServlet.java
package com.example.servlets;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.*;
import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;

@WebServlet("/register")
public class RegistrationServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                           HttpServletResponse response)
        throws ServletException, IOException {

        // 1. Lấy dữ liệu từ form
        String fullName = request.getParameter("fullName");
        String studentId = request.getParameter("studentId");
        String emailStr = request.getParameter("email");
        String phoneStr = request.getParameter("phone");
        String majorStr = request.getParameter("major");
        String yearStr = request.getParameter("year");
```

```
// 2. Trim whitespace
if (fullName != null) fullName = fullName.trim();
if (studentId != null) studentId = studentId.trim();
if (emailStr != null) emailStr = emailStr.trim();
if (phoneStr != null) phoneStr = phoneStr.trim();

// 3. Validate dữ liệu
String error = validateInput(fullName, studentId, emailStr, phoneStr,
majorStr, yearStr);

if (error != null) {
    // Có lỗi → quay lại form
    request.setAttribute("error", error);
    request.setAttribute("fullName", fullName);
    request.setAttribute("studentId", studentId);
    request.setAttribute("email", emailStr);
    request.setAttribute("phone", phoneStr);
    request.getRequestDispatcher("index.jsp").forward(request, response);
    return;
}

// 4. Xử lý logic (lưu vào file)
try {
    saveToFile(fullName, studentId, emailStr, phoneStr, majorStr,
yearStr);

    // 5. Set attributes cho JSP
    request.setAttribute("success", true);
    request.setAttribute("fullName", fullName);
    request.setAttribute("studentId", studentId);
    request.setAttribute("email", emailStr);
    request.setAttribute("phone", phoneStr);
    request.setAttribute("major", majorStr);
    request.setAttribute("year", yearStr);
    request.setAttribute("registrationTime", LocalDateTime.now()
        .format(DateTimeFormatter.ofPattern("dd/MM/yyyy HH:mm:ss")));

    // 6. Forward đến result JSP
    request.getRequestDispatcher("result.jsp").forward(request, response);
} catch (Exception e) {
    request.setAttribute("error", "System error: " + e.getMessage());
    request.getRequestDispatcher("error.jsp").forward(request, response);
}

}

/**
 * Validate input
 */
private String validateInput(String fullName, String studentId, String email,
String phone, String major, String year) {

    // Kiểm tra tên
    if (fullName == null || fullName.isEmpty()) {
```

```
        return "Full name is required";
    }
    if (fullName.length() < 3) {
        return "Full name must be at least 3 characters";
    }
    if (fullName.length() > 50) {
        return "Full name must be less than 50 characters";
    }

    // Kiểm tra student ID
    if (studentId == null || studentId.isEmpty()) {
        return "Student ID is required";
    }
    if (!studentId.matches("[0-9]{8}$")) {
        return "Student ID must be 8 digits";
    }

    // Kiểm tra email
    if (email == null || email.isEmpty()) {
        return "Email is required";
    }
    if (!email.matches("[A-Za-z0-9+_.-]+@(.+)$")) {
        return "Invalid email format";
    }

    // Kiểm tra phone
    if (phone == null || phone.isEmpty()) {
        return "Phone is required";
    }
    if (!phone.matches("[0-9]{10,11}$")) {
        return "Phone must be 10-11 digits";
    }

    // Kiểm tra major
    if (major == null || major.isEmpty()) {
        return "Please select a major";
    }

    // Kiểm tra year
    if (year == null || year.isEmpty()) {
        return "Please select graduation year";
    }

    return null; // Không có lỗi
}

/**
 * Lưu dữ liệu vào file
 */
private void saveToFile(String fullName, String studentId, String email,
                        String phone, String major, String year) throws
IOException {

    String logDir = getServletContext().getRealPath("/logs/");
```

```

File dir = new File(logDir);
if (!dir.exists()) {
    dir.mkdirs();
}

String timestamp = LocalDateTime.now()
    .format(DateTimeFormatter.ofPattern("yyyy-MM-dd_HH-mm-ss"));
String filename = logDir + "registration_" + timestamp + ".txt";

try (PrintWriter writer = new PrintWriter(new FileWriter(filename))) {
    writer.println("=== Student Registration ===");
    writer.println("Date: " + LocalDateTime.now()
        .format(DateTimeFormatter.ofPattern("dd/MM/yyyy HH:mm:ss")));
    writer.println();
    writer.println("Full Name: " + fullName);
    writer.println("Student ID: " + studentId);
    writer.println("Email: " + email);
    writer.println("Phone: " + phone);
    writer.println("Major: " + major);
    writer.println("Graduation Year: " + year);
    writer.println();
    writer.println("Status: Successfully registered");
}

System.out.println("Registration saved to: " + filename);
}

@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    // Redirect GET requests to form
    response.sendRedirect("index.jsp");
}
}

```

14.4 Code - index.jsp (Form)

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>

<!DOCTYPE html>
<html lang="vi">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Registration</title>
    <style>
        * {
            margin: 0;

```

```
padding: 0;
box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 20px;
}

.container {
  background: white;
  border-radius: 10px;
  box-shadow: 0 10px 40px rgba(0,0,0,0.2);
  width: 100%;
  max-width: 500px;
  padding: 40px;
}

h1 {
  color: #333;
  margin-bottom: 30px;
  text-align: center;
  font-size: 28px;
}

.error-message {
  background-color: #f8d7da;
  border: 1px solid #f5c6cb;
  color: #721c24;
  padding: 12px;
  border-radius: 5px;
  margin-bottom: 20px;
}

.form-group {
  margin-bottom: 20px;
}

label {
  display: block;
  margin-bottom: 8px;
  color: #333;
  font-weight: 600;
  font-size: 14px;
}

input[type="text"],
input[type="email"],
input[type="tel"],
```

```

    select {
        width: 100%;
        padding: 12px;
        border: 2px solid #e0e0e0;
        border-radius: 5px;
        font-size: 14px;
        transition: border-color 0.3s;
        font-family: Arial, sans-serif;
    }

    input[type="text"]:focus,
    input[type="email"]:focus,
    input[type="tel"]:focus,
    select:focus {
        outline: none;
        border-color: #667eea;
        box-shadow: 0 0 5px rgba(102, 126, 234, 0.3);
    }

    button {
        width: 100%;
        padding: 14px;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white;
        border: none;
        border-radius: 5px;
        font-size: 16px;
        font-weight: 600;
        cursor: pointer;
        transition: transform 0.2s, box-shadow 0.2s;
    }

    button:hover {
        transform: translateY(-2px);
        box-shadow: 0 5px 20px rgba(102, 126, 234, 0.4);
    }

    button:active {
        transform: translateY(0);
    }

    .help-text {
        font-size: 12px;
        color: #666;
        margin-top: 5px;
    }
}
</style>
</head>
<body>
    <div class="container">
        <h1>📖 Student Registration</h1>

        <% if (request.getAttribute("error") != null) { %>
            <div class="error-message">

```

```
<strong>✖ Error:</strong> <%= request.getAttribute("error") %>
</div>
<% } %>

<form method="POST" action="register">
  <!-- Full Name -->
  <div class="form-group">
    <label for="fullName">Full Name *</label>
    <input
      type="text"
      id="fullName"
      name="fullName"
      placeholder="E.g., Nguyễn Văn Nhân"
      value="<%= request.getAttribute("fullName") != null ?
request.getAttribute("fullName") : "" %>"
      required />
    <div class="help-text">Minimum 3 characters, maximum 50
characters</div>
  </div>

  <!-- Student ID -->
  <div class="form-group">
    <label for="studentId">Student ID *</label>
    <input
      type="text"
      id="studentId"
      name="studentId"
      placeholder="E.g., 12345678"
      value="<%= request.getAttribute("studentId") != null ?
request.getAttribute("studentId") : "" %>"
      required />
    <div class="help-text">8 digits only</div>
  </div>

  <!-- Email -->
  <div class="form-group">
    <label for="email">Email *</label>
    <input
      type="email"
      id="email"
      name="email"
      placeholder="your.email@example.com"
      value="<%= request.getAttribute("email") != null ?
request.getAttribute("email") : "" %>"
      required />
  </div>

  <!-- Phone -->
  <div class="form-group">
    <label for="phone">Phone Number *</label>
    <input
      type="tel"
      id="phone"
      name="phone"
```

```

        placeholder="09XXXXXXXX"
        value="<%= request.getAttribute("phone") != null ?
request.getAttribute("phone") : "" %>"
        required />
        <div class="help-text">10-11 digits</div>
    </div>

    <!-- Major -->
    <div class="form-group">
        <label for="major">Major *</label>
        <select id="major" name="major" required>
            <option value="">-- Select Major --</option>
            <option value="Computer Science">Computer Science</option>
            <option value="Software Engineering">Software
Engineering</option>
            <option value="Information Technology">Information
Technology</option>
            <option value="Data Science">Data Science</option>
            <option value="Business">Business</option>
        </select>
    </div>

    <!-- Graduation Year -->
    <div class="form-group">
        <label for="year">Expected Graduation Year *</label>
        <select id="year" name="year" required>
            <option value="">-- Select Year --</option>
            <option value="2025">2025</option>
            <option value="2026">2026</option>
            <option value="2027">2027</option>
            <option value="2028">2028</option>
        </select>
    </div>

    <button type="submit">☑ Register</button>
</form>
</div>
</body>
</html>

```

14.5 Code - result.jsp (Kết Quả)

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-
8" %>

<!DOCTYPE html>
<html lang="vi">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

```

```
<title>Registration Success</title>
<style>
  body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 20px;
  }

  .container {
    background: white;
    border-radius: 10px;
    box-shadow: 0 10px 40px rgba(0,0,0,0.2);
    width: 100%;
    max-width: 600px;
    padding: 40px;
    text-align: center;
  }

  .success-icon {
    font-size: 60px;
    margin-bottom: 20px;
  }

  h1 {
    color: #28a745;
    margin-bottom: 10px;
    font-size: 32px;
  }

  .subtitle {
    color: #666;
    margin-bottom: 30px;
    font-size: 16px;
  }

  .info-box {
    background: #f8f9fa;
    border-left: 4px solid #667eea;
    padding: 20px;
    margin: 20px 0;
    text-align: left;
    border-radius: 5px;
  }

  .info-row {
    display: flex;
    justify-content: space-between;
    padding: 10px 0;
    border-bottom: 1px solid #e0e0e0;
  }
```

```
.info-row:last-child {
  border-bottom: none;
}

.info-label {
  font-weight: 600;
  color: #333;
  min-width: 150px;
}

.info-value {
  color: #667eea;
  font-weight: 500;
}

.button-group {
  display: flex;
  gap: 10px;
  margin-top: 30px;
  justify-content: center;
}

button, a {
  padding: 12px 30px;
  border: none;
  border-radius: 5px;
  font-size: 16px;
  font-weight: 600;
  cursor: pointer;
  text-decoration: none;
  display: inline-block;
  transition: all 0.3s;
}

.btn-primary {
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  color: white;
}

.btn-primary:hover {
  transform: translateY(-2px);
  box-shadow: 0 5px 20px rgba(102, 126, 234, 0.4);
}

.btn-secondary {
  background: #e0e0e0;
  color: #333;
}

.btn-secondary:hover {
  background: #d0d0d0;
}
</style>
```

```

</head>
<body>
  <div class="container">
    <div class="success-icon">☑</div>

    <h1>Registration Successful!</h1>
    <p class="subtitle">Your registration has been processed successfully</p>

    <div class="info-box">
      <div class="info-row">
        <span class="info-label">Full Name:</span>
        <span class="info-value"><%= request.getAttribute("fullName") %>
      </span>
    </div>

    <div class="info-row">
      <span class="info-label">Student ID:</span>
      <span class="info-value"><%= request.getAttribute("studentId") %>
    </span>
    </div>

    <div class="info-row">
      <span class="info-label">Email:</span>
      <span class="info-value"><%= request.getAttribute("email") %>
    </span>
    </div>

    <div class="info-row">
      <span class="info-label">Phone:</span>
      <span class="info-value"><%= request.getAttribute("phone") %>
    </span>
    </div>

    <div class="info-row">
      <span class="info-label">Major:</span>
      <span class="info-value"><%= request.getAttribute("major") %>
    </span>
    </div>

    <div class="info-row">
      <span class="info-label">Graduation Year:</span>
      <span class="info-value"><%= request.getAttribute("year") %>
    </span>
    </div>

    <div class="info-row">
      <span class="info-label">Registration Time:</span>
      <span class="info-value"><%=
request.getAttribute("registrationTime") %></span>
    </div>
  </div>

  <p style="color: #666; margin-top: 20px;">
    ☑ Confirmation email has been sent to your email address.<br>

```

🎓 You will receive further instructions soon.

```
</p>

<div class="button-group">
  <a href="index.jsp" class="btn-primary">Register Another Student</a>
  <a href="javascript:window.print()" class="btn-secondary">Print</a>
</div>
</div>
</body>
</html>
```

14.6 web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app version="4.0"
  xmlns="http://xmlns.jcp.org/xml/ns/javaee"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
    http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd">

  <display-name>Student Registration System</display-name>

  <welcome-file-list>
    <welcome-file>index.jsp</welcome-file>
  </welcome-file-list>

</web-app>
```

14.7 Deploy & Chạy

Bước 1: Build WAR

```
mkdir -p build/StudentRegistration/WEB-INF/classes
javac -cp "C:\apache-tomcat-10.1.x\lib\*" ^
  -d build/StudentRegistration/WEB-INF/classes ^
  src/main/java/com/example/servlets/RegistrationServlet.java

copy src/main/webapp/WEB-INF/web.xml build/StudentRegistration/WEB-INF\
copy src/main/webapp/*.jsp build/StudentRegistration\

cd build
jar cvf StudentRegistration.war StudentRegistration/
```

Bước 2: Deploy

Copy `StudentRegistration.war` vào `C:\apache-tomcat-10.1.x\webapps\`

Bước 3: Truy cập

`http://localhost:8080/StudentRegistration/`

Hoàn thành Tuần 2!

Bạn đã học:

1. ☒ JSP Introduction (Directive, Scriptlet, Expression, Declaration)
2. ☒ JSP Lifecycle
3. ☒ 9 Implicit Objects
4. ☒ JSP Include (Page vs Action)
5. ☒ Expression Language (EL)
6. ☒ JSTL Core Tags
7. ☒ Form Handling
8. ☒ Mini Project - Student Registration

Bước tiếp theo: Tuần 3 - MVC Architecture

Tài liệu Tham Khảo

- Oracle JSP/Servlet Documentation
- Apache Tomcat Guide
- Java EE Platform
- JSTL Tutorial

Tài liệu được biên soạn bởi: Senior Java Developer Ngày cập nhật: 2025 Phiên bản: 1.0